

CIS 566 – Software Architecture

and Design patterns

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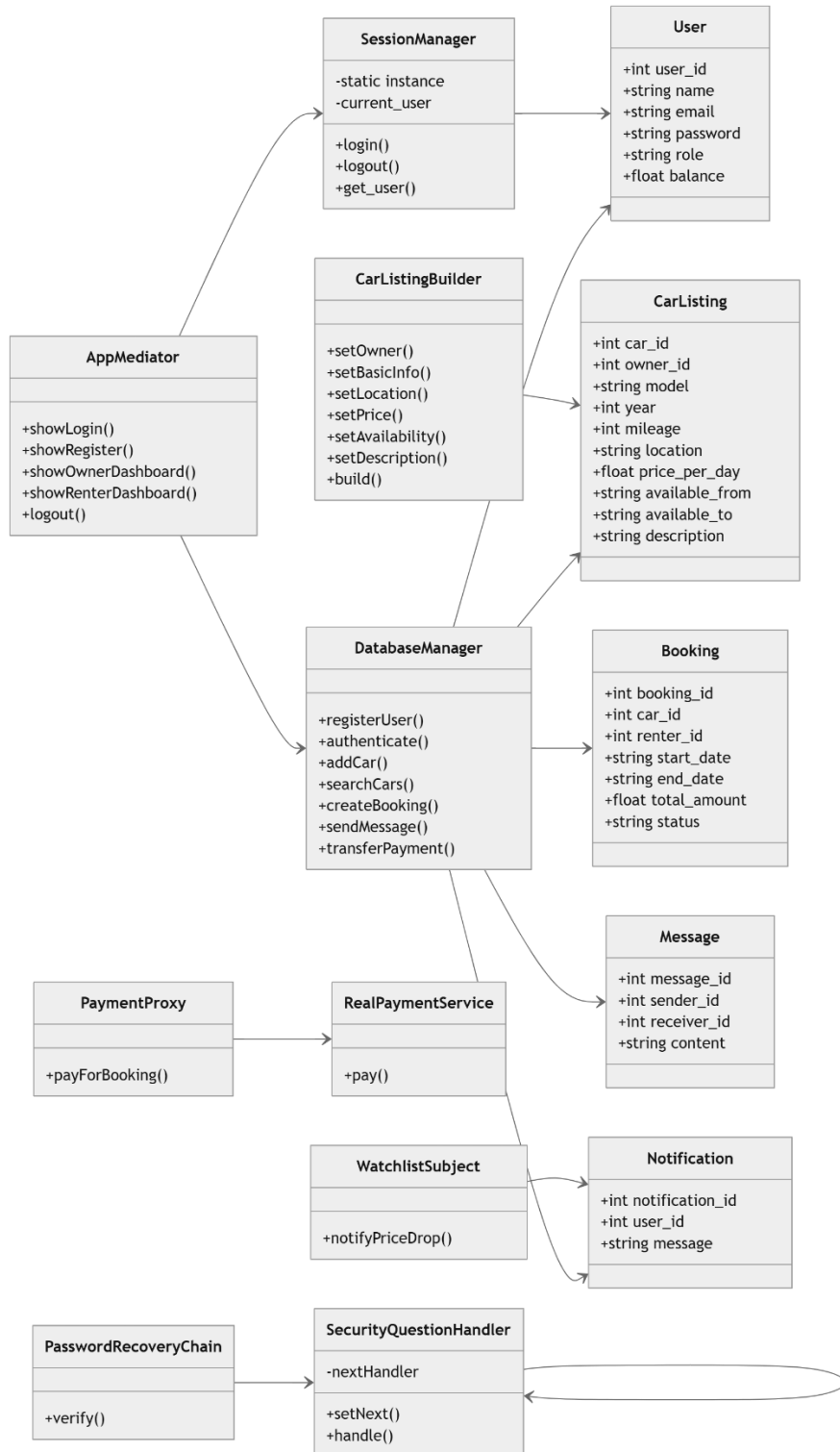
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Term Project

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UML Class Diagram



Database Schema

```
-- DriveShare Database Schema

CREATE TABLE users (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  name TEXT NOT NULL,
  email TEXT UNIQUE NOT NULL,
  password TEXT NOT NULL,
  role TEXT NOT NULL CHECK(role IN ('owner', 'renter')),
  balance REAL DEFAULT 1000.0,
  q1 TEXT NOT NULL,
  a1 TEXT NOT NULL,
  q2 TEXT NOT NULL,
  a2 TEXT NOT NULL,
  q3 TEXT NOT NULL,
  a3 TEXT NOT NULL
);

CREATE TABLE cars (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  owner_id INTEGER NOT NULL,
  model TEXT NOT NULL,
  year INTEGER NOT NULL,
  mileage INTEGER NOT NULL,
  location TEXT NOT NULL,
  price_per_day REAL NOT NULL,
  available_from TEXT NOT NULL,
  available_to TEXT NOT NULL,
  description TEXT,
  FOREIGN KEY(owner_id) REFERENCES users(id)
);

CREATE TABLE bookings (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  car_id INTEGER NOT NULL,
  renter_id INTEGER NOT NULL,
  start_date TEXT NOT NULL,
  end_date TEXT NOT NULL,
  total_amount REAL NOT NULL,
  status TEXT DEFAULT 'confirmed',
  paid INTEGER DEFAULT 0,
  FOREIGN KEY(car_id) REFERENCES cars(id),
  FOREIGN KEY(renter_id) REFERENCES users(id)
);
```

```
CREATE TABLE watchlists (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    renter_id INTEGER NOT NULL,  
    car_id INTEGER NOT NULL,  
    target_price REAL,  
    notify_when_available INTEGER DEFAULT 1,  
    FOREIGN KEY(renter_id) REFERENCES users(id),  
    FOREIGN KEY(car_id) REFERENCES cars(id)  
);  
  
CREATE TABLE messages (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    sender_id INTEGER NOT NULL,  
    receiver_id INTEGER NOT NULL,  
    content TEXT NOT NULL,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY(sender_id) REFERENCES users(id),  
    FOREIGN KEY(receiver_id) REFERENCES users(id)  
);  
  
CREATE TABLE notifications (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    user_id INTEGER NOT NULL,  
    message TEXT NOT NULL,  
    is_read INTEGER DEFAULT 0,  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
    FOREIGN KEY(user_id) REFERENCES users(id)  
);
```

Class Diagram Description

The class diagram illustrates the main components of the DriveShare system and their relationships. Core classes include `User`, `CarListing`, `Booking`, `Message`, and `Notification`, which represent the primary entities of the platform. Additional classes implement the required design patterns, including `SessionManager` (Singleton), `CarListingBuilder` (Builder), `WatchlistSubject` (Observer), `AppMediator` (Mediator), `PaymentProxy` (Proxy), and `PasswordRecoveryChain` (Chain of Responsibility). Together, these classes support user authentication, vehicle management, bookings, messaging, notifications, and payment processing.

Database Schema Description

The DriveShare database consists of six main tables: `users`, `cars`, `bookings`, `watchlists`, `messages`, and `notifications`. The `users` table stores account information and security questions. The `cars` table stores vehicle listings created by owners. The `bookings` table manages reservations and payment status. The `watchlists` table supports renter notifications for desired vehicles. The `messages` table stores communication between users, while the `notifications` table stores system-generated alerts. These tables work together to support the complete car rental workflow.

How To Run

The application was developed using Python, Tkinter, and SQLite. To execute the system, open a terminal in the project directory and run:

```
python main.py
```

The SQLite database is automatically created during the first execution. No external server, cloud service, or third-party payment provider is required.

For testing purposes, the following demo accounts are available:

Owner Account

Email: owner@test.com

Password: 123

Renter Account

Email: renters@test.com

Password: 123

These accounts can be used to quickly demonstrate vehicle listing, searching, booking, messaging, notifications, and payment functionality without creating new users.

GitHub Repository Link

<https://github.com/ReemJ23/566-term-project>

Presentation Video Link

<https://youtu.be/OEdoCuF68Go>